GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Positive Regulation Of Cytokine Producti	0.08873829	262	8.673e−07	4.672e−03	CD80:27 SYK:48 CD86:55 IL7:138 TNF:189 NLRP9:217
Cytoplasmic Translation (GO:0002181)	0.22112506	39	1.793e−06	4.828e−03	RPS14:17 ETF1:59 RPS15A:79 RACK1:109 RPS27:207 RPL19:252
Gene Expression (GO:0010467)	0.08599387	205	2.363e−05	2.546e−02	RPS14:17 RPS15A:79 CASP3:112 MAGOH:176 RPS27:207 RNASEL:227
Peptide Biosynthetic Process (GO:0043043	0.12423791	99	2.009e−05	2.546e−02	RPS14:17 GGTLC3:68 RPS15A:79 GGTLC1:194 RPS27:207 DMD:244
Proton Motive Force−Driven Mitochondrial	0.18036291	47	1.915e−05	2.546e−02	NDUFS5:43 NDUFA11:242 NDUFA12:363 NDUFB5:753 NDUFS3:808 NDUFB3:851
Oxidative Phosphorylation (GO:0006119)	0.16271677	54	3.590e−05	3.224e−02	NDUFS5:43 NDUFA11:242 NDUFA12:363 NDUFB5:753 NDUFS3:808 NDUFB3:851
Nervous System Development (GO:0007399)	−0.06252316	363	4.866e−05	3.744e−02	FZD9:41 RBFOX3:73 RITA1:117 LRCH4:142 TMEM223:145 EPHB1:276
Aerobic Electron Transport Chain (GO:001	0.15915949	52	7.284e−05	3.934e−02	NDUFS5:43 COX5A:455 COX5B:535 NDUFB5:753 NDUFS3:808 NDUFB3:851
Mitochondrial ATP Synthesis Coupled Elec	0.16085264	52	6.097e−05	3.934e−02	NDUFS5:43 NDUFA12:363 COX5A:455 COX5B:535 NDUFB5:753 NDUFS3:808
Organic Hydroxy Compound Transport (GO:0	0.19382903	35	7.302e−05	3.934e−02	SLC44A2:425 SLC44A1:477 ATP8B1:508 ABCC11:538 SLCO1A2:678 SLCO1B1:684
Leukotriene D4 Biosynthetic Process (GO:	0.45342092	6	1.199e−04	5.754e−02	GGTLC3:68 GGTLC1:194 GGT5:320 GGT6:744 GGTLC2:1109 GGT7:1695
Negative Regulation Of Gene Expression (	0.06771009	273	1.282e−04	5.754e−02	ENG:22 ATP2B4:54 RACK1:109 TASOR:135 SLFN14:187 TNF:189
Positive Regulation Of Macromolecule Met	0.06469852	296	1.408e−04	5.833e−02	DNMT3B:18 FGF21:104 CLU:116 CALCR:144 TNF:189 ACTC1:210
Leukotriene D4 Metabolic Process (GO:190	0.37817833	8	2.123e−04	7.925e−02	GGTLC3:68 GGTLC1:194 GGT5:320 GGT6:744 GGTLC2:1109 GGT1:1121
Regulation Of Synaptic Transmission, Glu	−0.15586869	47	2.207e−04	7.925e−02	HCN1:160 NLGN2:287 CCR2:596 NRXN1:607 NLGN3:733 GRIN2D:943
Mitochondrial Electron Transport, NADH T	0.19041138	31	2.449e−04	8.246e−02	NDUFS5:43 NDUFB5:753 NDUFS3:808 NDUFB3:851 NDUFA3:991 NDUFA5:1076
Regulation Of Chemokine Production (GO:0	0.17095358	38	2.679e−04	8.490e−02	IL7:138 TNF:189 IFNG:226 TLR7:279 IL33:297 EIF2AK2:305
RNA Processing (GO:0006396)	0.09501374	122	2.988e−04	8.665e−02	SNRPD1:31 CDK12:159 SRSF1:163 MAGOH:176 RNASEL:227 U2AF1:231
Positive Regulation Of Tumor Necrosis Fa	0.12866120	66	3.056e−04	8.665e−02	SYK:48 CD86:55 CLU:116 IFNG:226 IL33:297 TLR9:443
Proton Motive Force−Driven ATP Synthesis	0.14560129	51	3.257e−04	8.772e−02	NDUFS5:43 NDUFA11:242 NDUFA12:363 NDUFB5:753 NDUFS3:808 NDUFB3:851
Defense Response To Symbiont (GO:0140546	0.10667803	93	3.861e−04	9.903e−02	SERINC3:169 NLRP9:217 RNASEL:227 STAT2:235 TLR7:279 EIF2AK2:305
Macrophage Differentiation (GO:0030225)	0.22569397	20	4.771e−04	1.117e−01	IFNG:226 CDC42:228.5 IL33:297 IL31RA:402 LIF:670 CASP10:994
Positive Regulation Of Chemokine Product	0.15597622	42	4.736e−04	1.117e−01	SYK:48 IL7:138 TNF:189 IFNG:226 TLR7:279 IL33:297
Bile Acid And Bile Salt Transport (GO:00	0.25092320	16	5.120e−04	1.120e−01	ATP8B1:508 ABCC11:538 SLCO1A2:678 SLCO1B1:684 SLC51A:865 ABCD3:1015
Positive Regulation Of Nervous System De	−0.15306545	43	5.196e−04	1.120e−01	EPHB1:276 NLGN2:287 SERPINF1:304 ZNF335:457 NRXN1:607 SLITRK1:727
Positive Regulation Of NF−kappaB Transcr	0.08902680	126	5.754e−04	1.192e−01	CLU:116 TNF:189 AKR1B1:271 CARD14:275 EIF2AK2:305 IL18RAP:339
Aerobic Respiration (GO:0009060)	0.13575775	53	6.360e−04	1.224e−01	NDUFS5:43 NDUFA11:242 NDUFA12:363 NDUFB5:753 NDUFS3:808 NDUFB3:851
Myeloid Leukocyte Differentiation (GO:00	0.12897566	59	6.193e−04	1.224e−01	IFNG:226 CDC42:228.5 IL33:297 IL4:322 IL31RA:402 GLO1:445
Dendritic Cell Differentiation (GO:00970	0.24582426	16	6.646e−04	1.235e−01	CLU:116 FLT3:157 IL4:322 LYN:1411 CCR7:2895 BATF:3011.5
Histone Ubiquitination (GO:0016574)	0.34559514	8	7.122e−04	1.279e−01	UBE2B:23 UBR2:80 WAC:1817 RNF40:2232 UBE2A:2966 UBE2N:3418
Sphingolipid Biosynthetic Process (GO:00	0.11787975	68	7.868e−04	1.367e−01	ST8SIA5:96 ELOVL7:151 ST8SIA1:196 HACD2:200 SPTLC3:269 PLPP3:280
Steroid Hormone Biosynthetic Process (GO	0.24111488	16	8.421e−04	1.418e−01	HSD17B2:62 CYP17A1:145 HSD17B11:270 CYP19A1:506 HSD17B3:600 HSD3B2:815
Negative Regulation Of Rho Protein Signa	0.25641479	14	8.957e−04	1.419e−01	DYNLL1:6 RASIP1:1136 KANK1:1143 MYOC:1732 RIPOR2:1764 FLCN:2034
Positive Regulation Of Leukocyte Cell−Ce	0.17558557	30	8.776e−04	1.419e−01	TNF:189 PRKCQ:466 ETS1:1131 CCDC88B:1315 RHOA:1367 FCHO1:1868
Positive Regulation Of Tyrosine Phosphor	0.13925232	47	9.659e−04	1.487e−01	FLT3:157 TNF:189 IFNG:226 PARP14:311 IL4:322 IL31RA:402
Positive Regulation Of Tumor Necrosis Fa	0.11908814	62	1.197e−03	1.791e−01	SYK:48 CLU:116 IFNG:226 IL33:297 TLR9:443 ORM2:626
Positive Regulation Of Interleukin−6 Pro	0.11807126	62	1.319e−03	1.920e−01	SYK:48 TNF:189 IFNG:226 TLR7:279 IL33:297 TLR8:334
Negative Regulation Of Type I Interferon	0.17377900	28	1.465e−03	2.077e−01	KLHL22:359 ITCH:569 REL:703 ATG5:1438 RELB:1606 GPATCH3:1663
Calcium−Mediated Signaling Using Intrace	0.21855018	17	1.814e−03	2.334e−01	CLU:116 GPR143:170 VCAM1:370 DMTN:1583 CCL20:2641 PTGFR:3062
Leukotriene Biosynthetic Process (GO:001	0.22576204	16	1.772e−03	2.334e−01	GGTLC3:68 GGTLC1:194 GGT5:320 GGT6:744 MGST3:823 GGTLC2:1109

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_INFLUENZA_INFECTION	0.16934998	82	1.175e−07	2.351e−04	RPL23:7 RPS14:17 RPS15A:79 RPS27:207 POLR2J:246 RPL19:252
REACTOME_INFECTIOUS_DISEASE	0.06166290	671	6.459e−08	2.351e−04	ANO5:1 DYNLL1:6 RPL23:7 TUBB8:13 TUBB8B:15 RPS14:17
REACTOME_SARS_COV_1_HOST_INTERACTIONS	0.20580913	57	7.786e−08	2.351e−04	RPS14:17 RPS15A:79 UBE2I:94 RPS27:207 TLR7:279 RPS7:374
FLECHNER_BIOPSY_KIDNEY_TRANSPLANT_OK_VS_	0.07413660	432	1.449e−07	2.351e−04	DYNLL1:6 SULT1A2:9 TSPAN8:28 ETF1:59 SCFD1:92 RACK1:109
REACTOME_CELLULAR_RESPONSES_TO_STIMULI	0.06250254	589	2.626e−07	3.407e−04	H3−3A:3 DYNLL1:6 RPL23:7 TUBB8:13 TUBB8B:15 RPS14:17
MARSON_BOUND_BY_FOXP3_STIMULATED	0.04856589	854	1.826e−06	1.626e−03	UBE2D1:37 TMEM50A:52 ETF1:59 FBXW12:67 H4C15:103 MYL4:120
REACTOME_SARS_COV_1_MODULATES_HOST_TRANS	0.35623684	15	1.781e−06	1.626e−03	RPS14:17 RPS15A:79 RPS27:207 RPS7:374 RPS26:481 RPS15:597
PUJANA_BRCA1_PCC_NETWORK	0.04009058	1283	2.005e−06	1.626e−03	RPL23:7 TRA2B:8 HIRA:29 SNRPD1:31 MVK:33 HK2:42
WP_NONALCOHOLIC_FATTY_LIVER_DISEASE	0.12297172	123	2.545e−06	1.834e−03	NDUFS5:43 CASP3:112 BID:179 TNF:189 CDC42:228.5 NDUFA11:242
REACTOME_EUKARYOTIC_TRANSLATION_ELONGATI	0.21074641	41	3.043e−06	1.855e−03	RPL23:7 RPS14:17 RPS15A:79 EEF1B2:121 RPS27:207 RPL19:252
REACTOME_SARS_COV_1_INFECTION	0.13997249	93	3.146e−06	1.855e−03	RPS14:17 RPS15A:79 UBE2I:94 ST3GAL2:177 RPS27:207 TLR7:279
MULLIGHAN_NPM1_SIGNATURE_3_UP	0.08137843	275	3.641e−06	1.968e−03	UBE2K:58 UBR2:80 SERINC3:169 ST3GAL2:177 ACP2:286 SLC46A3:365
WP_NETWORK_MAP_OF_SARSCOV2_SIGNALING_PAT	0.10270060	165	5.517e−06	2.722e−03	FGG:30 CASP3:112 APOD:114 CXCL13:126 TNFSF10:127 IL7:138
SCHLOSSER_SERUM_RESPONSE_DN	0.05654773	556	5.875e−06	2.722e−03	TRA2B:8 UBR2:80 UBE2G1:90 PPP3CC:107 RBBP4:115 EEF1B2:121
KEGG_HUNTINGTONS_DISEASE	0.11038698	128	1.653e−05	7.147e−03	NDUFS5:43 CASP3:112 TFAM:185 POLR2J:246 SLC25A4:308 COX5A:455
REACTOME_MRNA_SPLICING_MINOR_PATHWAY	0.24516817	25	2.209e−05	8.495e−03	SNRPD1:31 SF3B5:122 SF3B6:128 SRSF1:163 POLR2J:246 SNRPD3:534
HSIAO_HOUSEKEEPING_GENES	0.07646127	261	2.226e−05	8.495e−03	H3−3A:3 DYNLL1:6 RPL23:7 RPS14:17 RPS15A:79 JUND:88
KEGG_RIBOSOME	0.20052473	37	2.444e−05	8.809e−03	RPL23:7 RPS15A:79 RPS27:207 RPL19:252 RPS7:374 MRPL13:446
YAO_TEMPORAL_RESPONSE_TO_PROGESTERONE_CL	0.10861709	125	2.805e−05	9.579e−03	GATAD2A:81 SRSF1:163 NARS1:230 DTYMK:349 RARS1:353 CARM1:452
REACTOME_SPERM_MOTILITY_AND_TAXES	0.42619112	8	2.986e−05	9.685e−03	CATSPERB:106 CATSPER3:403 KCNU1:958 CATSPERD:1007 CATSPERG:1423 CATSPER4:1441
REACTOME_GLUCOSE_METABOLISM	0.13321519	81	3.448e−05	1.065e−02	HK2:42 SEH1L:309 GNPDA2:494 GOT1:510 HK1:523 GOT2:546
DARWICHE_SQUAMOUS_CELL_CARCINOMA_UP	0.11060910	117	3.661e−05	1.079e−02	DYNLL1:6 ENG:22 EEF1B2:121 SRSF1:163 BID:179 ICAM1:431
SEIDEN_ONCOGENESIS_BY_MET	0.14004251	72	4.022e−05	1.134e−02	TSPAN8:28 ADAM9:65 HAT1:260 UGDH:565 HIGD1A:840 EPS8:1044
MARTORIATI_MDM4_TARGETS_FETAL_LIVER_DN	0.05758122	432	4.417e−05	1.146e−02	FGG:30 TNPO3:47 BNC1:70 COL1A1:123 MAB21L2:139 SRSF1:163
WP_MIRNA_ROLE_IN_IMMUNE_RESPONSE_IN_SEPS	0.21216185	31	4.359e−05	1.146e−02	TNF:189 TLR7:279 TLR8:334 VCAM1:370 ICAM1:431 REL:703
BURTON_ADIPOGENESIS_6	0.09094916	163	6.336e−05	1.522e−02	HSD11B1:225 CA3:330 S100A8:444 COX5B:535 ORM2:626 ME1:641
KEGG_GLUTATHIONE_METABOLISM	0.19028553	37	6.218e−05	1.522e−02	GGT5:320 OPLAH:329 IDH1:612 GGT6:744 MGST3:823 GSTO1:850
DIAZ_CHRONIC_MYELOGENOUS_LEUKEMIA_UP	0.03515991	1149	7.362e−05	1.706e−02	R3HDM4:2 SULT1A2:9 DNMT3B:18 UBE2B:23 HIRA:29 SNRPD1:31
REACTOME_ACTIVATION_OF_THE_MRNA_UPON_BIN	0.21467407	28	8.453e−05	1.891e−02	RPS14:17 RPS15A:79 RPS27:207 RPS7:374 RPS26:481 RPS15:597
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	0.04878556	555	9.402e−05	1.967e−02	CD80:27 UBE2D1:37 SYK:48 CD86:55 TNFSF8:111 CASP3:112
REACTOME_METABOLISM_OF_LIPIDS	0.04528549	650	9.222e−05	1.967e−02	MVK:33 HSD17B2:62 PI4K2A:73 CYP51A1:75 UBE2I:94 RAB14:113
REACTOME_GLUTATHIONE_CONJUGATION	0.22349256	25	1.099e−04	2.228e−02	GGT5:320 OPLAH:329 GSTT2B:723 GGT6:744 CHAC1:750 MGST3:823
REACTOME_SARS_COV_2_MODULATES_HOST_TRANS	0.21835992	26	1.165e−04	2.289e−02	RPS14:17 SNRPD1:31 RPS15A:79 RPS27:207 RPS7:374 RPS26:481
DARWICHE_PAPILLOMA_RISK_LOW_UP	0.09949041	124	1.326e−04	2.458e−02	DYNLL1:6 ENG:22 EEF1B2:121 SRSF1:163 LRIF1:278 COL20A1:325
TIEN_INTESTINE_PROBIOTICS_24HR_UP	0.05223673	462	1.292e−04	2.458e−02	DYNLL1:6 DNMT3B:18 SNRPD1:31 UBE2D1:37 TMEM50A:52 ETF1:59
REACTOME_SARS_COV_INFECTIONS	0.06238650	312	1.590e−04	2.761e−02	ANO5:1 RPS14:17 SNRPD1:31 SYK:48 RPS15A:79 GATAD2A:81
LEE_LIVER_CANCER_SURVIVAL_DN	0.09581378	131	1.557e−04	2.761e−02	DYNLL1:6 UBE2D1:37 CALM2:222 OPLAH:329 H4C14:340 RPS7:374
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.20600036	28	1.617e−04	2.761e−02	HSD17B2:62 CYP17A1:145 HSD11B1:225 CYP19A1:506 COMT:547 HSD17B3:600
REACTOME_DRUG_ADME	0.13676973	63	1.752e−04	2.914e−02	SULT1A3:21 SULT1A4:24 HSD11B1:225 GGT5:320 SLC29A2:665 SLCO1A2:678
KEGG_PHENYLALANINE_METABOLISM	0.26236093	17	1.804e−04	2.926e−02	PRDX6:479 GOT1:510 GOT2:546 PAH:889 TAT:1042 MIF:1333

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MYELOPROLIFERATIVE DISORDER, CHRONIC, WI	0.21711512	38	3.678e−06	1.657e−02	TNF:189 CXADR:223 IFNG:226 CXCR2:251 ICAM1:431 COMT:547
Hyperglycemia	0.06684943	416	3.472e−06	1.657e−02	TRPV1:19 HK2:42 JUND:88 FGF21:104 CASP3:112 TNFSF10:127
Lymphoid leukemia	0.10056624	174	5.071e−06	1.657e−02	RPS14:17 SYK:48 PRMT5:150 FLT3:157 TNF:189 IFNG:226
Malignant tumor of colon	0.03518899	1540	7.922e−06	1.693e−02	PROM1:4 TRA2B:8 DNMT3B:18 CD80:27 HK2:42 CEMIP:45
Pneumonia	0.06586816	393	8.635e−06	1.693e−02	DNMT3B:18 FGG:30 CEMIP:45 SYK:48 CD86:55 ICOS:76
Inflammatory dermatosis	0.11197252	130	1.090e−05	1.782e−02	FLT3:157 TNF:189 IFNG:226 CXCR2:251 GRN:265 TLR7:279
Ichthyosis, X−Linked	0.13944686	80	1.662e−05	1.811e−02	CASP3:112 TNFSF10:127 ATF1:174 TNF:189 DMD:244 MIB1:387
Osteoporosis, Senile	0.16593220	57	1.496e−05	1.811e−02	PGLS:87 COL1A1:123 CALCR:144 HSD11B1:225 U2AF1:231 CYP19A1:506
Dengue Fever	0.11344480	123	1.452e−05	1.811e−02	SYK:48 TNFSF10:127 LTA:182 TNF:189 HSPA5:197 IFNG:226
Thyroid associated opthalmopathies	0.14479881	73	1.933e−05	1.895e−02	CD80:27 CD86:55 LTA:182 TNF:189 ST8SIA1:196 IFNG:226
Thromboangiitis Obliterans	0.19405813	40	2.192e−05	1.954e−02	TNF:189 ST8SIA1:196 IFNG:226 VCAM1:370 ICAM1:431 RBM45:512
Gingival Diseases	0.12745011	92	2.461e−05	2.011e−02	CD80:27 CLU:116 SNAP25:143 TNF:189 IFNG:226 IL4:322
Autoimmune Diseases	0.04411395	790	3.299e−05	2.423e−02	RPL23:7 CD80:27 TNPO3:47 AFP:49 CD86:55 GGTLC3:68
Small head	−0.05419296	508	3.460e−05	2.423e−02	BCOR:3 MATN4:23 PCNT:33 RB1:64 ALDH18A1:97 ATRX:102
Malaria, Cerebral	0.14075296	69	5.382e−05	2.989e−02	LTA:182 TNF:189 IFNG:226 TLR7:279 IL33:297 IL4:322
Myeloid Leukemia, Chronic	0.04725737	636	5.873e−05	2.989e−02	PROM1:4 DNMT3B:18 CD80:27 SCPEP1:46 SYK:48 CD86:55
Adult Hepatocellular Carcinoma	0.12713940	83	6.376e−05	2.989e−02	DNMT3B:18 ENG:22 SAV1:105 SRSF1:163 AKR1B1:271 GJB2:427
Chronic inflammatory disorder	0.19272626	36	6.345e−05	2.989e−02	TNF:189 CXCR2:251 CARD14:275 IL33:297 TLR9:443 IL1RN:537
Colitis	0.05300791	499	5.923e−05	2.989e−02	TRPV1:19 CD86:55 ICOS:76 ROS1:83 COL2A1:108 TNFSF8:111
Nonalcoholic Steatohepatitis	0.08966150	168	6.403e−05	2.989e−02	DNMT3B:18 FGF21:104 CASP3:112 COL1A1:123 TNFSF10:127 CYP17A1:145
Vasculitis	0.12114650	93	5.548e−05	2.989e−02	MVK:33 CXCL13:126 TNF:189 IFNG:226 AKR1B1:271 VCAM1:370
Microcephaly	−0.08372857	191	6.996e−05	3.118e−02	PCNT:33 ATRX:102 UBE3B:113 MED17:132 CENPJ:139 MYO16:166
Anemia	0.05871318	390	7.786e−05	3.319e−02	RPS14:17 DNMT3B:18 ENG:22 ATP2B4:54 CASP3:112 TNFSF10:127
Adult−Onset Still Disease	0.26679597	18	8.929e−05	3.521e−02	CLU:116 TNF:189 IFNG:226 IL4:322 ICAM1:431 MAP4K3:693
Dermatitis	0.09170263	154	8.977e−05	3.521e−02	FLT3:157 TNF:189 IFNG:226 CXCR2:251 GRN:265 TLR7:279
Colon Carcinoma	0.03018822	1610	9.404e−05	3.546e−02	PROM1:4 TRA2B:8 DNMT3B:18 ENG:22 CD80:27 TSPAN8:28
Eosinophilia−Myalgia Syndrome	0.45252727	6	1.237e−04	4.492e−02	COL1A1:123 IFNG:226 RBM45:512 TGFB1:846 CAT:963 FN1:1480
Congenital dyserythropoietic anemia, typ	0.32121791	11	2.254e−04	5.140e−02	MAN2A1:352 AICDA:662 MGAT2:892 TFRC:916 SEC23B:1867 KLF1:2352
Melanoma, B16	0.10755548	101	1.926e−04	5.140e−02	GGTLC3:68 TNF:189 GGTLC1:194 CALM2:222 CXADR:223 IFNG:226
Benign recurrent intrahepatic cholestasi	0.43705365	6	2.094e−04	5.140e−02	GGTLC3:68 GGTLC1:194 ATP8B1:508 GGT2:849 GGT1:1121 ABCB11:2800
cervical cancer	0.04167793	696	2.181e−04	5.140e−02	DNMT3B:18 ENG:22 UBE2B:23 CD80:27 HK2:42 FSHR:53
Chronic active hepatitis	0.14028608	59	1.965e−04	5.140e−02	CASP3:112 CYP17A1:145 IFNG:226 EIF2AK2:305 IL1RN:537 AIRE:571
Crohn Disease	0.04351120	665	1.581e−04	5.140e−02	CD86:55 JUND:88 TNFSF8:111 CASP3:112 COL1A1:123 CMC1:175
Everted lower lip vermilion	−0.12776248	71	2.007e−04	5.140e−02	ATRX:102 GTF2IRD1:201 EDA:478 WDR35:608 EDARADD:874 PSAT1:973
Liver carcinoma	0.02398301	2597	1.771e−04	5.140e−02	PROM1:4 DYNLL1:6 IQGAP3:11 DNMT3B:18 SLC22A16:20 SULT1A3:21
Mental impairment	−0.08756312	151	2.124e−04	5.140e−02	PCNT:33 ALG2:67 NAGS:81 CLCF1:86 CENPJ:139 COL5A1:189
Neoplasm Metastasis	0.02267921	2913	2.225e−04	5.140e−02	PROM1:4 DYNLL1:6 SULT1A2:9 IQGAP3:11 PPP4C:12 DNMT3B:18
Nephrotic Syndrome	0.08705696	153	2.103e−04	5.140e−02	CD80:27 LMX1B:63 CLU:116 COL1A1:123 TNF:189 IL4:322
Periodontal Diseases	0.10143234	112	2.147e−04	5.140e−02	CD80:27 CLU:116 IL7:138 SNAP25:143 TNF:189 IFNG:226
Post−Traumatic Osteoporosis	0.14722408	53	2.118e−04	5.140e−02	PGLS:87 COL1A1:123 CALCR:144 U2AF1:231 CYP19A1:506 VCL:683

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0001845 abnormal inflammatory respons	0.03734070	475	7.307e−03	2.818e−01	TRPV1:19 CD80:27 JUND:88 CASP3:112 TNFSF10:127 BID:179
MP0008770 decreased survivor rate	0.06477405	146	7.555e−03	2.818e−01	DNMT3B:18 SAV1:105 MAGOH:176 SYNE1:201 PLEC:221 IFNG:226
MP0000266 abnormal heart morphology	0.03295895	651	6.571e−03	2.818e−01	DNMT3B:18 ENG:22 HIRA:29 DYNC2H1:50 PNN:72 JUND:88
MP0002135 abnormal kidney morphology	0.03840815	420	9.050e−03	2.818e−01	TRPV1:19 BCKDK:35 HSD17B2:62 LMX1B:63 BCAM:136 SLC12A1:192
MP0002136 abnormal kidney physiology	0.05088098	267	5.074e−03	2.818e−01	TRPV1:19 LMX1B:63 COL2A1:108 SLC12A1:192 IFNG:226 JUNB:267
MP0009643 abnormal urine homeostasis	0.05702732	218	4.336e−03	2.818e−01	LMX1B:63 SLC12A1:192 IFNG:226 JUNB:267 AQP1:272 MAN2A1:352
MP0001819 abnormal immune cell	0.03090108	822	5.003e−03	2.818e−01	PPP4C:12 CD80:27 FGG:30 CD86:55 ICOS:76 JUND:88
MP0005000 abnormal immune tolerance	0.05320205	237	5.617e−03	2.818e−01	CD80:27 CD86:55 ICOS:76 TNFSF10:127 LTA:182 TNF:189
MP0001984 abnormal olfaction	−0.20865097	20	1.257e−03	2.818e−01	APOE:89 BBS1:377 PDE1C:628 NLGN3:733 CNGA4:1672 NAV2:1768
MP0005408 hypopigmentation	0.17564870	21	5.399e−03	2.818e−01	MAGOH:176 PAH:889 LYST:955 RAB27A:1237 SLC31A1:1353 ATOX1:2062
MP0010307 abnormal tumor latency	−0.14688249	27	8.376e−03	2.818e−01	FOXO3:174 CTSZ:271 PTCH1:327 NBN:339 BUB1:976 BRCA2:1725
MP0005165 increased susceptibility to	0.09646403	61	9.493e−03	2.818e−01	TRPV1:19 CAPN8:40 LTA:182 TNF:189 GRN:265 CD2AP:819
MP0000026 abnormal inner ear	−0.05818846	180	7.924e−03	2.818e−01	RB1:64 APOE:89 FOXI1:228 DUOX2:248 BBS1:377 APAF1:426
MP0003119 abnormal digestive system	−0.11111947	48	7.958e−03	2.818e−01	HGS:60 NODAL:235 RAPGEF2:335 NKX3−2:450 MCM4:546 IGF2BP1:554
MP0002420 abnormal adaptive immunity	0.03304622	818	2.737e−03	2.818e−01	PPP4C:12 CD80:27 FGG:30 CD86:55 JUND:88 CASP3:112
MP0006054 spinal hemorrhage	0.29974617	8	3.343e−03	2.818e−01	COL1A1:123 PSEN1:722 FLI1:974 C1GALT1:999 FBLN1:2092 SPHK1:5583
MP0000343 altered response to	−0.10438466	50	1.096e−02	3.063e−01	MAPKAPK2:66 APOE:89 F13A1:111 CCR2:596 LEPR:652 TIMP1:958
MP0005330 cardiomyopathy	0.08953279	66	1.232e−02	3.251e−01	JUND:88 TFAM:185 DMD:244 IRX4:292 CALR:438 MYOZ2:922
MP0000001 mammalian phenotype	0.01947199	2790	1.742e−02	4.248e−01	PROM1:4 TRA2B:8 PPP4C:12 DNMT3B:18 TRPV1:19 ENG:22
MP0005025 abnormal response to	0.03844867	339	1.788e−02	4.248e−01	FGG:30 ICOS:76 TNFSF10:127 FLT3:157 LTA:182 TNF:189
MP0002060 abnormal skin morphology	0.07930586	72	2.066e−02	4.673e−01	PLEC:221 JUNB:267 COL17A1:291 GJB2:427 IL1R1:430 DSE:530
MP0002452 abnormal antigen presenting	0.02960636	538	2.459e−02	4.866e−01	CD80:27 CD86:55 ICOS:76 CASP3:112 COL1A1:123 CXCL13:126
MP0008057 abnormal DNA replication	0.20599482	10	2.419e−02	4.866e−01	LIG1:155 REL:703 PAXIP1:746 DICER1:2693 GABPA:3849 CDC7:5302
MP0003950 abnormal plasma membrane	0.21956724	9	2.263e−02	4.866e−01	DMD:244 CDK5RAP2:347 CAV3:1454 PIGA:1477 GLRX:1677 TJP2:4195
MP0010234 abnormal vibrissa follicle	0.32171275	4	2.589e−02	4.920e−01	ST14:240 CHUK:2398 PKP3:3462 SPINK5:4393 NA NA
MP0003787 abnormal imprinting	−0.22173004	8	2.996e−02	5.474e−01	RB1:64 ATRX:102 ARID4B:747 RASGRF1:2001 ARID4A:2959 CDKN1C:6664
MP0005389 reproductive system phenotype	0.06927754	64	5.646e−02	5.756e−01	DLG1:204 COL17A1:291 CYP19A1:506 RXFP1:776 BAD:787 LGR4:818
MP0002166 altered tumor susceptibility	0.11817819	22	5.539e−02	5.756e−01	RAG1:236 ADAM15:644 LYST:955 PSME1:1116 PSME2:1525 BMPR1A:1991
MP0000647 abnormal sebaceous gland	−0.08220194	43	6.301e−02	5.756e−01	APOE:89 PTCH1:327 EDA:478 LEPR:652 EDARADD:874 DLX3:1504
MP0001784 abnormal fluid regulation	0.03154467	375	4.180e−02	5.756e−01	ENG:22 HIRA:29 PNN:72 JUND:88 SAV1:105 CLU:116
MP0002006 tumorigenesis	0.10377918	29	5.358e−02	5.756e−01	RAG1:236 ADAM15:644 LYST:955 PSME1:1116 PSME2:1525 IL1A:1735
MP0001727 abnormal embryo implantation	0.11441470	25	4.809e−02	5.756e−01	PCYT1A:483 ARNTL:559 LIF:670 MRC1:816 VEZT:1160 PTGS2:2185
MP0003136 yellow coat color	−0.24666667	6	3.648e−02	5.756e−01	EDA:478 OCA2:1214 ADAMTS20:3883 MC1R:5036 MITF:6495 EDNRB:6666
MP0000003 abnormal adipose tissue	0.09612548	32	6.045e−02	5.756e−01	BCKDK:35 FGF21:104 COL1A1:123 CARM1:452 E2F1:997 LEP:1449
MP0009764 decreased sensitivity to	0.07521921	57	5.048e−02	5.756e−01	BID:179 LTA:182 TNF:189 CXCR2:251 IL33:297 ACE:555
MP0001756 abnormal urination	0.06339339	79	5.283e−02	5.756e−01	TRPV1:19 PI4K2A:73 SLC12A1:192 AQP1:272 CD63:404 ACE:555
MP0003436 decreased susceptibility to	0.08973616	38	5.629e−02	5.756e−01	LTA:182 TNF:189 CXCR2:251 IL1R1:430 CTSS:830 TNFSF13:1241
MP0010771 integument phenotype	0.04260532	175	5.511e−02	5.756e−01	TRPV1:19 SCN11A:89 SAV1:105 DDB2:130 PLEC:221 IFNG:226
MP0003948 abnormal gas homeostasis	−0.03400112	264	6.251e−02	5.756e−01	SNTA1:15 ECEL1:18 RB1:64 STAT5A:77 TGFB3:148 TCF21:164
MP0002067 abnormal sensory capabilities	−0.03959074	238	3.893e−02	5.756e−01	RB1:64 INPP4A:129 EN1:168 MYOG:177 GTF2IRD1:201 MYO5A:210

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
breast	0.139431337	14	7.104e−02	8.590e−01	LMX1B:63 TNN:124 FABP4:1250 GSC:1366 LEP:1449 THRSP:1658
Brodmann.area.24	−0.058587236	29	2.755e−01	8.590e−01	KCNS1:464 DDN:653 IDS:872 DRD5:925 TMEM271:1225 SLC17A7:1233
caudate.nucleus	−0.056562083	26	3.187e−01	8.590e−01	LRRC10B:589 PPP1R1B:890 VAX1:1878 WIF1:2027 MAPK4:3212 CHAT:3227
cerebellar.hemisphere	−0.024662186	459	7.712e−02	8.590e−01	TAFA3:7 BRICD5:11 ABLIM1:22 KCNC1:27 WSCD2:36 CCNP:39
cerebellum	−0.020630670	354	1.905e−01	8.590e−01	TAFA3:7 KCNC1:27 WSCD2:36 CCNP:39 BSN:61 RBFOX3:73
cerebral.cortex	−0.030740008	103	2.835e−01	8.590e−01	OPRD1:162 CASKIN1:281 KCNS1:464 CHRM1:485 CABP1:539 NCALD:582
cortex.kidney	0.028403343	110	3.061e−01	8.590e−01	SLC12A1:192 SLC39A5:255 SLCO4C1:294 PDZK1:532 SLC28A1:688 SLC4A9:730
EBV.lymphocyte	−0.009550772	652	4.217e−01	8.590e−01	GINS2:38 RFX5:45 PRC1:90 ZMAT3:106 H2AC13:141 POLD1:152
endocervix	−0.054104872	70	1.188e−01	8.590e−01	CILP:481 OSR2:489 TACR1:512 SFRP4:667 SP5:780 NBL1:1165
esophagus.mucosa	0.017237686	213	3.910e−01	8.590e−01	IQANK1:56 BNC1:70 FAM83H:133 KRT78:162 EVPL:298 GJB2:427
esophagus.muscularis.mucosa	0.071232397	26	2.092e−01	8.590e−01	JPH2:454 FAM83D:604 PRUNE2:832 NKX6−1:1808 SYNM:2083 TPM1:2830
greater.omentum	0.038455882	32	4.522e−01	8.590e−01	BNC1:70 GFPT2:731 HBEGF:971 FABP4:1250 LEP:1449 THRSP:1658
leg.skin	0.015939451	209	4.319e−01	8.590e−01	BNC1:70 POF1B:152 KRT78:162 SPTLC3:269 CARD14:275 COL17A1:291
liver	0.023938640	321	1.470e−01	8.590e−01	FGG:30 AFP:49 HSD17B2:62 FGF21:104 GRB14:149 HSD11B1:225
lung	0.021511791	139	3.846e−01	8.590e−01	ROS1:83 WNT2:184 GGTLC1:194 MSR1:429 ICAM1:431 SUSD2:463
nucleus.accumbens	0.033704468	93	2.635e−01	8.590e−01	DACH2:26 HTR1D:99 FAT3:398 NECAB2:529 FBXL16:674 GABRA5:915
pancreas	0.068215507	93	2.363e−02	8.590e−01	FBXW12:67 CELA3A:85 CATSPERB:106 SLC39A5:255 SLC17A4:663 TEX11:718
peyers.patch	0.022888556	162	3.187e−01	8.590e−01	SULT1A2:9 SLC39A5:255 SI:257 MOGAT3:293 TMEM37:399 TMIGD1:417
pituitary.gland	−0.023736242	202	2.497e−01	8.590e−01	DGKK:80 FSHB:119 CRHR2:122 VMO1:302 VWDE:427 SCN3B:526
putamen	−0.109709592	14	1.555e−01	8.590e−01	LRRC10B:589 GBX2:1081 RGS9:1539 PDE10A:2430 DIPK1C:3777 GPR88:3793
sigmoid.colon	0.030187260	50	4.613e−01	8.590e−01	MAB21L2:139 ACTG2:228.5 BAG2:245 TACR2:434 HMCN2:1827 WNT9A:1950
spleen	−0.013799429	290	4.257e−01	8.590e−01	C1QA:62 FHOD1:197 CD6:204 CCL24:323 RASGRP2:328 GSDMD:356
suprapubic.skin	0.018451510	181	3.962e−01	8.590e−01	BNC1:70 TLCD3A:95 KRT78:162 CXADR:223 CARD14:275 COL17A1:291
tibial.artery	−0.023231509	101	4.221e−01	8.590e−01	AVPR1A:14 SOST:400 KCNE4:436 SORBS2:763 FLNA:821 ADARB1:968
tibial.nerve	−0.038624075	183	7.419e−02	8.590e−01	SLC9A3:13 IL9:443 CDH19:515 LEPR:652 FOXD2:717 GAS2L3:751
transverse.colon	0.020814124	144	3.920e−01	8.590e−01	TSPAN8:28 ST14:240 SLC39A5:255 MOGAT3:293 TMIGD1:417 SLC44A4:489
uterus	−0.051344747	53	1.971e−01	8.590e−01	CHRDL2:557 TGFB1I1:886 SHISAL1:997 GAS1:1319 HOXA11:1691 ADAMTS19:2076
vagina	0.039330842	89	2.017e−01	8.590e−01	BNC1:70 KRT78:162 EVPL:298 GJB2:427 RHCG:460 CASP14:503
eye.development	−0.051423577	57	1.805e−01	8.590e−01	RORB:784 CHRDL1:947 FZD7:985 NBL1:1165 MSX1:1244 GAS1:1319
adrenal.gland	0.012175242	107	6.652e−01	8.812e−01	CYP17A1:145 AKR1B1:271 GML:306 KLHL4:373 KLF14:511 IDH1:612
amygdala	0.022329279	8	8.269e−01	8.812e−01	PCDH15:132 PTPRZ1:935 BCAN:1500 NEUROD2:4911 SLC38A8:10489 NEUROD6:12973
aorta	−0.014848870	151	5.320e−01	8.812e−01	COL21A1:94 EFEMP1:215 NIBAN1:265 SOST:400 CAVIN3:497 SMOC2:519
atrium.auricle	−0.010400299	89	7.356e−01	8.812e−01	ASB10:116 LMOD2:290 MYOM3:297 ITGB1BP2:316 SYNPO2L:417.5 NPPA:495
bladder	0.022690557	23	7.067e−01	8.812e−01	SLC14A1:757 SEMA3E:1341 CLEC3A:1809 ACER2:1930 PLA2G4A:1999 ID1:3027
Brodmann.area.9	−0.006740148	162	7.691e−01	8.812e−01	FRMPD4:20 CBLN4:112 HCN1:160 PHYHIPL:348 AJAP1:409 KCNS1:464
C1.spinal.cord	0.005106095	171	8.194e−01	8.812e−01	ELOVL7:151 PAQR6:186 SLC44A1:477 TBC1D12:647 SLCO1A2:678 CLDN11:875
coronary.artery	−0.028192180	14	7.151e−01	8.812e−01	PCOLCE2:217 SOST:400 CCDC190:953 TIMP1:958 CCN3:3021 HEYL:6008
ectocervix	−0.029043632	25	6.156e−01	8.812e−01	PRLR:84 COL5A1:189 TRHDE:1167 SOCS2:2674 CPXM1:2769 TSKU:3587
esophagogastric.junction	0.033319900	8	7.442e−01	8.812e−01	NKX6−1:1808 ADCY5:3146 GADL1:5564 HOXA4:7904 F2RL2:7959 CHRM3:8716
fallopian.tube	0.012053484	48	7.732e−01	8.812e−01	ADGRG4:36 GATA5:215 PI15:689 TCF23:1034 RARRES1:1138 HOXA3:1386